Your Health, Our Mission

# HOSPITAL MANAGMENT SYSTEM

database project presentation

GROUP # 8

# GROUP MEMBERS

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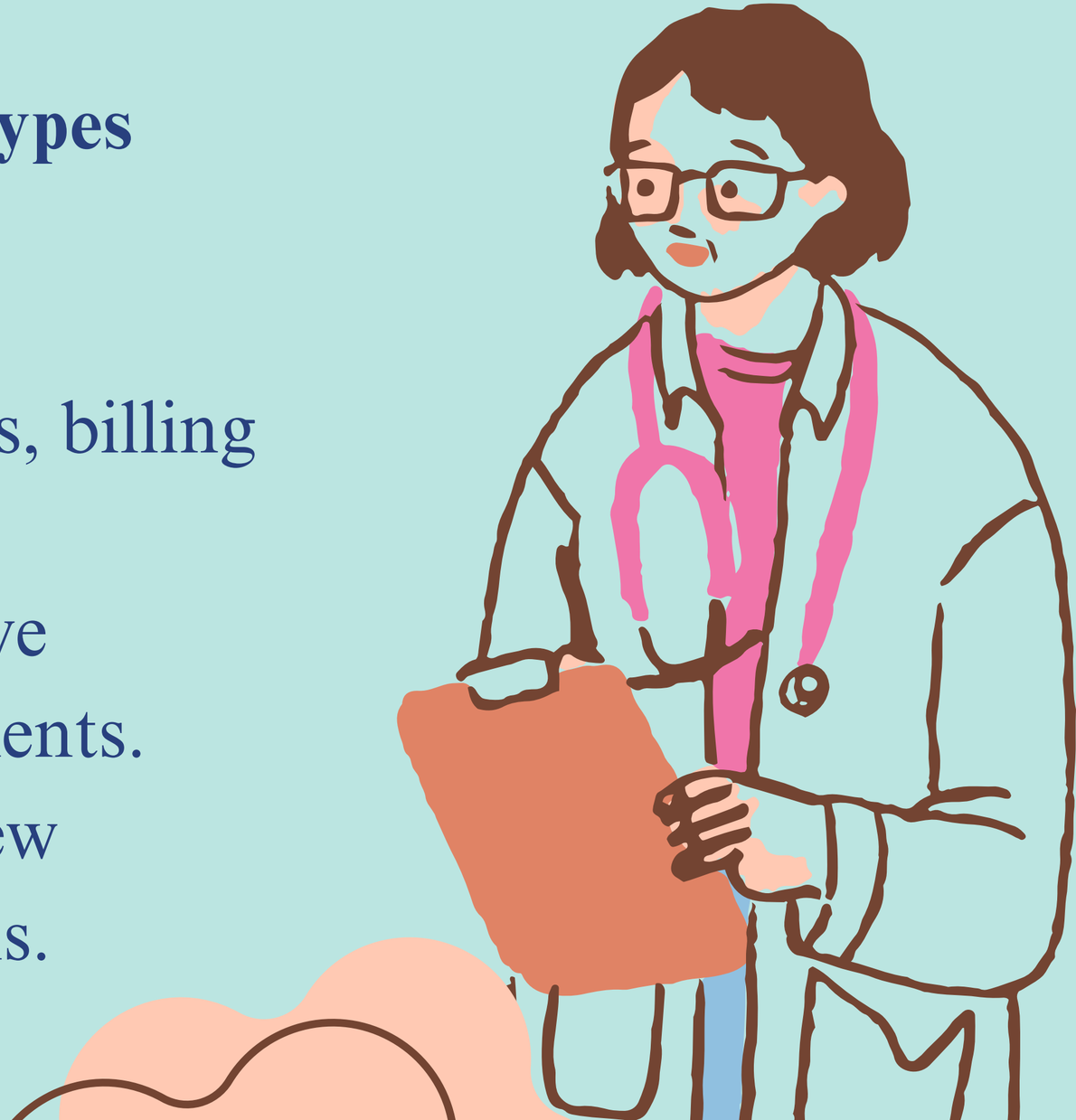
# PROJECT OVERVIEW

The Hospital Management System (HMS) is designed to:

- Keep track of **patients' records, appointments, consultations, billing.**
- Provide **secure, real-time access** to medical data for doctors, staff, and patients.
- Automate workflows like **appointment scheduling.**
- Ensure **data security**, **stock monitoring**, and integration with external systems.

**The system supports different user types such as:**

- **Admin Staff:** Handle registrations, billing and reports.
- **Doctors:** View patient history, give consultations, and update treatments.
- **Patients:** Book appointments, view consultation history, and pay bills.



# ER DIAGRAM

## Main Entities and Relationships:

### PATIENT

### DOCTOR

### APPOINTMENT

## INVENTORY

## CONSULTATION

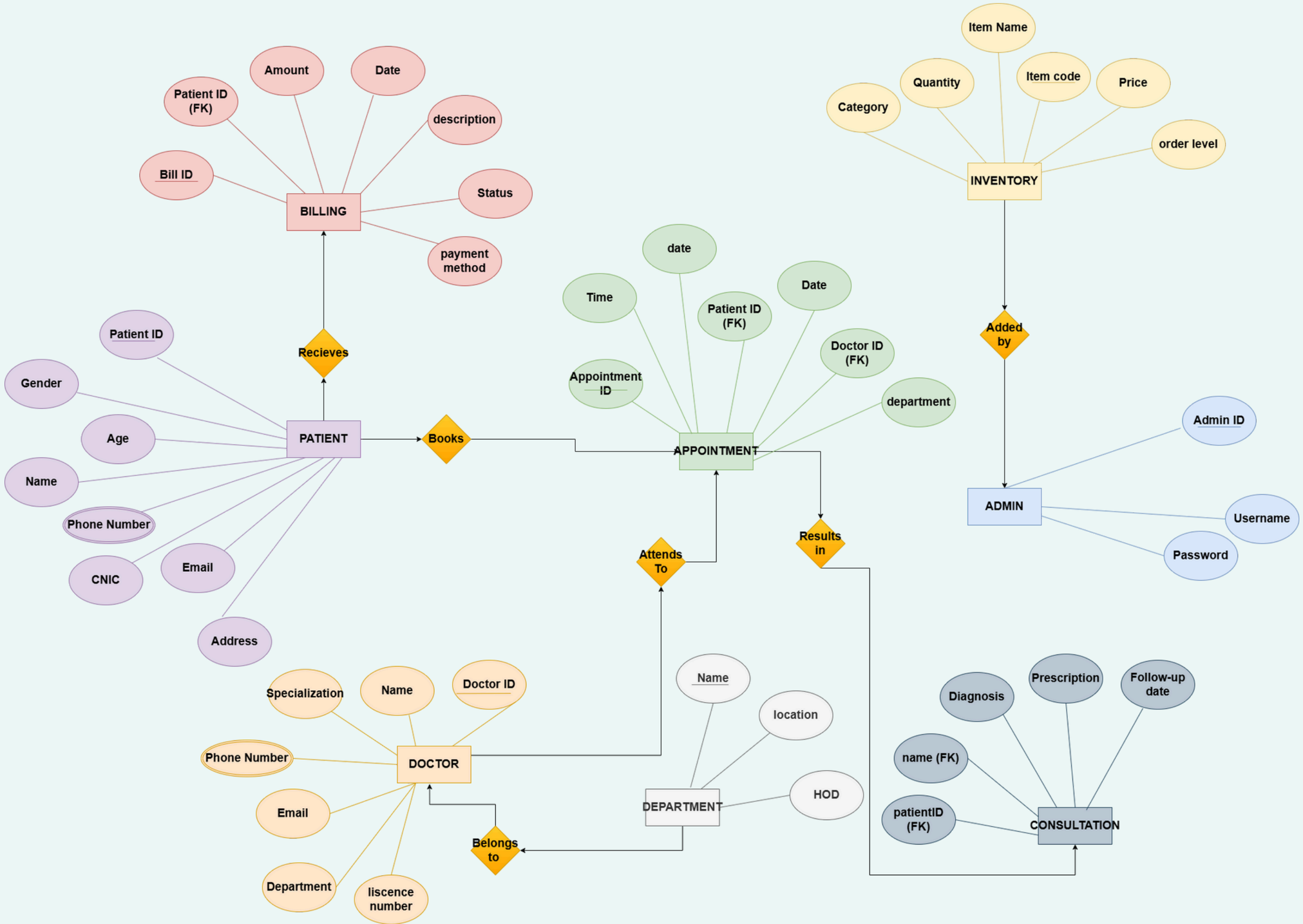
## BILLING

DEPARTMENT

ADMIN

# KEY RELATIONSHIPS: ER DIAGRAM

- A **doctor** can attend many **appointments**.
- A **patient** can book many **appointments**.
- Each **appointment** results in one **consultation**.
- Each **doctor** belongs to one **department**.
- An **admin** can add things in **inventory**.
- A **patient** can receive multiple **billing** records.


Amount
Date
Patient ID (FK)
description
Bill ID
Status
BILLING
payment method
Item Name
Quantity
Item code
Price
Category
order level
INVENTORY
date
Time
Date
Patient ID (FK)
Doctor ID (FK)
Appointment ID
department
Added by
Patient ID
Recieves
Gender
Age
PATIENT
Books
APPOINTMENT
Admin ID
Name
ADMIN
Phone Number
Username
Email
Results in
Attends To
CNIC
Password
Address
Name
Specialization
Name
Doctor ID
location
Diagnosis
Prescription
Follow-up date
Phone Number
DOCTOR
name (FK)
DEPARTMENT
HOD
Email
patientID (FK)
CONSULTATION
Belongs to
Department
liscence number

| AdminID | Username | Password |
|---|---|---|

| AppointmentID | PatientID | DoctorID | AppointmentDate | Time | Added_by |
|---|---|---|---|---|---|

| BillID | PatientID | Amount | Date | Description | Status | PaymentMethod |
|---|---|---|---|---|---|---|

| name | PatientID | Diagnosis | Prescription | FollowupDate |
|---|---|---|---|---|

| Name | Location | HOD |
|---|---|---|

| DoctorID | Name | Specialization | PhoneNumber | Email | Password | Department | LicenceNumber |
|---|---|---|---|---|---|---|---|

| ItemID | ItemName | Category | Quantity | ItemCode | Price | OrderLevel |
|---|---|---|---|---|---|---|

| PatientID | Name | Age | Gender | Email | Password |
|---|---|---|---|---|---|

# FRONT-END PAGES

01. appointment

02 billing_payment

03. consultation

04. dashboard_admin

05. dashboard_doctor

06. dashboard_patient

07. doctor_profile

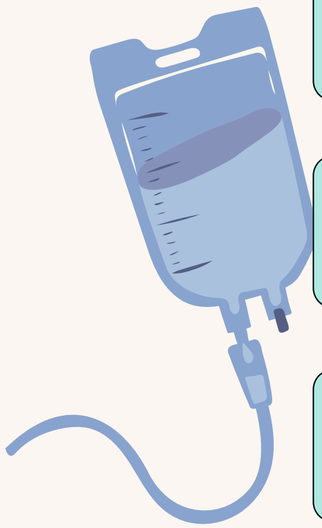
08. inventory_management

09. login

10. logout

11. patient_registration

12. reports_analytics



# HMS - Patient Portal

Dashboard | Book Appointment | My Appointments | My Consultations | Medical Records | Billing & Payment | Logout

## Quick Actions

+ Book New Appointment

View Medical Records

Request Prescription Refill

## Today's Appointments

Dr. Sarah Khan - Cardiology

Join Now | Details

# HMS - Doctor Portal

Dashboard | My Profile | Appointments | Consultations | Patient Records | Logout

8

View Appointments

5

View Schedule

3

View Queue

## Quick Actions

Start Consultation

View Patient Records

Write Prescription

# HMS - Admin Portal

Dashboard | Register Patient | Manage Doctors | Appointments | Inventory | Reports | Logout

View Patients

View Appointments

Manage Doctors

## Reports

7

View Reports

## Recent Activity

New patient registered 10 mins ago

# Rules Applied for Data Integrity and Coherence

## Primary Keys (PK)

Unique identifiers like **PatientID**, **DoctorID**, **AppointmentID**, etc.

## Foreign Keys (FK)

Ensures relational integrity between tables.

- **PatientID** in Appointment references **Patient table**.
- **DoctorID** in Appointment links to **Doctor**.

## NOT NULL Constraints

- Critical fields like patient name, consultation date, and billing amount cannot be empty.
- Unique Constraints
- Fields like Username for Admins must be unique to avoid duplicate login conflicts.

## ON DELETE CASCADE / UPDATE CASCADE

- When a doctor is deleted, all linked appointments and consultations are also removed (if cascade used).

## Validation Logic

- Inventory quantity can't be negative.
- Appointment cannot be made without selecting a doctor or room.
- Bills are marked "Paid" only after payment info is provided.

# THANK YOU